(a) Phenomenon

The phenomenon I am interested in modeling is the time required for a driver to find a legal street parking space in New York City.

(b) Importance of predicting this phenomenon

Predicting parking search time is important because it can reduce traffic congestion, fuel consumption, and driver frustration. A significant amount of urban traffic is caused by drivers searching for parking, so accurate predictions could help navigation systems recommend better decisions (e.g., whether to search for street parking or use a garage). It could also help city planners better understand parking demand and improve urban infrastructure.

(c) Response metric (y)

The response variable y is the time (in minutes) it takes for a driver to successfully find a street parking space.

(d) Measurement of the response

The response will be measured in minutes, starting from the moment a driver begins actively searching for parking until the moment they park in a street parking spot. This measurement can be obtained using GPS tracking data or driver-reported timestamps.

(e) Justification for measuring n observations

This response can be measured many times relatively easily because each parking attempt by a driver constitutes one observation. Data could be collected through GPS navigation apps, parking sensors, or surveys. Since thousands of drivers search for parking in NYC every day, it is feasible to obtain a large sample size n.

(f) Preliminary features / proxies

Since true causal drivers (like exact available spots or driver behavior) are not directly observable, the following proxy features can be used:

- Neighborhood/location
- Time of day

- Day of the week
- Population density or traffic level
- Number of parking spaces in the area
- Weather
- Parking restrictions (e.g., street cleaning schedules)
- Proximity to high-demand areas (restaurants, offices, events)
- Special events or holidays

These features are measurable and are likely to influence parking availability and search time.